

Probability – 1

Understanding Probability

Probability helps describe how likely something is to happen when the outcome is uncertain. Instead of giving a definite answer, it assigns a value that reflects chance. This value always lies between zero and one, where one extreme represents impossibility and the other represents certainty.

In everyday situations, probability is constantly at play. Whether it's a simple random process or a complex business prediction, it helps in making better judgments under uncertainty.

- Common situations where probability appears
 - Flipping a coin and observing the result
 - Throwing a dice and noting the number
 - Estimating whether a user will stop using a service

Different Approaches to Probability

Probability can be understood in more than one way depending on how information is obtained.

- Theoretical perspective
 - Based on all possible outcomes being known in advance
 - Works well in controlled scenarios like games of chance
 - Example: If outcomes are equally likely, each outcome shares the total chance equally
- Experimental perspective
 - Derived from real observations or past data
 - Useful in practical scenarios where outcomes are not perfectly predictable
 - Example: Estimating how often users click on an advertisement based on historical records

Basic Concepts

To work with probability, a few fundamental ideas are essential.

- Sample space
 - The complete set of all possible results of an experiment

- Example: When a dice is thrown, every number it can show forms the full set of outcomes
- Event
 - A specific result or a group of results from the sample space
 - Example: Getting an even number includes multiple possible outcomes from the full set

These concepts help in organizing and analyzing uncertain situations in a structured way.

Applications in Real-World Scenarios

Probability is widely used across industries to solve practical problems.

- Business and analytics applications
 - Estimating how many users will respond to a campaign
 - Predicting chances of product defects in production
 - Evaluating risk levels in insurance systems
 - Identifying suspicious or unusual financial activities

These applications show how probability supports smarter and data-driven decisions.

Calculating Probability

At its core, probability compares the number of desired outcomes to the total possible outcomes.

- General idea
 - Probability of an event depends on how many ways that event can occur relative to all possible outcomes
- Expression
 - Probability of an event = favorable cases divided by total possible cases
- Example
 - Consider a situation where an outcome is chosen randomly from a fixed set
 - If only one outcome matches the desired condition and there are six possible outcomes overall, the chance becomes one out of six
 - This can also be written as a fraction or converted into a decimal value for easier interpretation

This simple approach forms the basis for more advanced probability concepts used in data science and analytics.

Experiments in Probability

An experiment in probability refers to any process or action that leads to one or more outcomes. These outcomes may be predictable or uncertain depending on the nature of the experiment. Understanding the type of experiment helps in deciding how probability can be applied.

- Types of experiments
 - Deterministic experiments
 - Produce the same result every time under the same conditions
 - No uncertainty involved
 - Example: Heating water to a fixed temperature under identical conditions will always give the same result
 - Random experiments
 - Outcomes are not fixed and can differ even if conditions remain unchanged
 - Involves uncertainty and chance
 - Example: Flipping a coin multiple times may give different sequences of heads and tails

Sample Space

The sample space represents the complete collection of all possible outcomes of a random experiment. It acts as the foundation for analyzing probability because every event must belong to this set.

- Key idea
 - It includes every outcome that can possibly occur, without omission
- Examples
 - Rolling a die
 - All possible results form a set containing six values
 - Tossing two coins
 - Outcomes include every combination of heads and tails in sequence form

Counting Outcomes in Common Experiments

In many cases, instead of listing all outcomes, we calculate how many are possible using patterns.

- For coins
 - Each coin has two possible outcomes
 - When multiple coins are used, total outcomes grow exponentially
 - Total outcomes can be expressed as
 - 2 raised to the power of number of coins
 - Example
 - With three coins, total outcomes become $2 \times 2 \times 2$
- For dice
 - Each die has six possible outcomes
 - With multiple dice, outcomes increase similarly
 - Total outcomes can be written as
 - 6 raised to the power of number of dice
 - Example
 - With two dice, total outcomes become 6×6

Understanding Cards as an Experiment

A deck of playing cards is another common setup used in probability. It contains a fixed number of distinct outcomes.

- Structure of a standard deck
 - Total cards: 52
 - Divided into two broad categories
 - Red cards
 - Black cards
 - Each category contains two groups
 - Red category
 - Hearts

- Diamonds
- Black category
 - Spades
 - Clubs
- Each group contains cards starting from Ace, followed by numbered cards, and ending with face cards

Connecting Experiments and Probability

Probability becomes meaningful only when applied to experiments, especially those involving uncertainty. Every random experiment produces outcomes, and probability helps quantify how likely each outcome is within the complete set of possibilities.

- The connection
 - A random experiment generates outcomes
 - These outcomes together form the sample space
 - Probability assigns a likelihood to events within that space

For example, when an experiment has equally likely outcomes, each outcome shares the total chance evenly. This idea simplifies many basic calculations.

Events and Their Types

An event represents a condition or outcome of interest within an experiment. Events can vary in complexity depending on how many outcomes they include.

- Simple event
 - Contains only one outcome
 - Example: Getting a specific number when rolling a die
- Compound event
 - Includes more than one outcome
 - Example: Getting an even number when rolling a die
- Certain event
 - Covers the entire sample space

- Always occurs
- Impossible event
 - Contains no outcomes from the sample space
 - Cannot occur

Relationship Between Outcomes and Probability

The likelihood of an event depends directly on how many outcomes satisfy the condition compared to the total number of outcomes.

- General reasoning
 - More favorable outcomes lead to higher likelihood
 - Fewer favorable outcomes lead to lower likelihood
- Expression
 - Chance of an event = count of favorable outcomes divided by total outcomes
- Illustration
 - If an experiment has eight possible results and two meet the desired condition, the likelihood is two out of eight

Complementary Thinking

Sometimes it is easier to calculate the chance of an event by considering its opposite.

- Complement of an event
 - Represents all outcomes where the event does not occur
 - Total probability of all possible outcomes equals one
 - So, probability of an event plus probability of its complement equals one
- Example
 - If the chance of an event happening is known, subtracting it from one gives the chance of it not happening

This approach is especially useful when the event itself is complex but its opposite is simpler to calculate.

Combining Multiple Experiments

When more than one experiment is involved, the number of outcomes increases rapidly. Understanding how outcomes combine is important.

- Independent situations
 - Outcome of one does not affect the other
 - Total outcomes can be found by multiplying possibilities
- Example
 - Tossing a coin and rolling a die together
 - Total outcomes become the product of outcomes from each experiment

Important Probability Definitions

Basic Foundations

- Experiment
 - Any process that produces outcomes
 - Example: Drawing a card from a deck
- Random experiment
 - An experiment where the result cannot be predicted exactly
 - Example: Tossing a coin and observing the result
- Outcome
 - A single possible result
 - Example: Getting “heads” in a coin toss
- Sample space
 - The set of all possible outcomes
 - Example: For a coin $\rightarrow \{\text{Heads, Tails}\}$
- Event
 - A subset of the sample space
 - Example: Getting heads from the coin toss

Types of Events

- Simple event
 - Contains only one outcome
 - Example: Rolling a 4 on a die
- Compound event
 - Contains multiple outcomes
 - Example: Rolling an even number $\rightarrow \{2, 4, 6\}$
- Sure event
 - Includes all possible outcomes
 - Example: Getting a number between 1 and 6 on a die
- Impossible event
 - Cannot occur
 - Example: Rolling a 7 on a standard die
- Complementary event
 - Outcomes where the event does not occur
 - Example: If event is getting heads, complement is getting tails
- Mutually exclusive events
 - Cannot happen at the same time
 - Example: Getting 2 and 5 in a single die roll
- Exhaustive events
 - Together cover all outcomes
 - Example: Getting an even number or an odd number on a die

Probability Basics

- Probability of an event
 - Measures how likely an event is
 - Expression
 - $\text{Probability} = \text{favorable outcomes} \div \text{total outcomes}$

- Example
 - Rolling a die and getting a 2
 - Favorable outcomes = 1, total outcomes = 6
 - Probability = $1 \div 6$
- Range of probability
 - Values lie between zero and one
 - Example
 - Impossible event $\rightarrow 0$
 - Certain event $\rightarrow 1$
- Classical approach
 - Based on equally likely outcomes
 - Example: Probability of getting heads in a fair coin $\rightarrow 1 \div 2$
- Empirical approach
 - Based on observed data
 - Example: If 200 clicks occur in 1000 views, probability of click = $200 \div 1000$

Conditional Probability

- Conditional probability
 - Likelihood of an event given another event has already occurred
 - Expression
 - Probability of A given B = probability of both A and B $\div$ probability of B
 - Example
 - From cards, probability of drawing a king given that the card is a face card
 - Favorable cases = kings, reduced sample = face cards

Independence and Dependence

- Independent events
 - One event does not affect another
 - Example: Tossing a coin and rolling a die
- Dependent events
 - One event influences another
 - Example: Drawing two cards without replacement

Random Variables

- Random variable
 - Assigns numerical values to outcomes
 - Example: Number of heads in two coin tosses
- Discrete random variable
 - Takes countable values
 - Example: Number of customers entering a store
- Continuous random variable
 - Takes values over a range
 - Example: Height of a person

Probability Distribution

- Probability distribution
 - Shows how probabilities are assigned to values
- Discrete distribution
 - Example
 - Number of heads in two coin tosses with probabilities assigned to 0, 1, and 2
- Continuous distribution
 - Example
 - Distribution of heights in a population

Expectation and Variability

- Expected value
 - Average outcome over many trials
 - Example
 - For a fair die, average value = $(1+2+3+4+5+6) \div 6$
- Variance
 - Measures spread from the average
 - Example
 - If values are widely spread, variance is high
- Standard deviation
 - Square root of variance
 - Example
 - Used to measure consistency in data

Types of Probability

- Joint probability
 - Probability of two events happening together
 - Example: Getting heads and rolling a 3
- Marginal probability
 - Probability of a single event
 - Example: Probability of getting heads regardless of other events
- Conditional independence
 - Events become independent under a condition
 - Example: Two tests may be independent once a factor is fixed

Continuous Probability Concepts

- Probability density
 - Describes likelihood over a range
 - Example: Distribution of weights in a population
- Cumulative distribution
 - Probability up to a certain value
 - Example: Probability that height is less than a given value

Experiment and Outcome

- Experiment
 - Any process that generates results
 - Can be predictable or uncertain
 - Types
 - Deterministic
 - Always gives the same result under fixed conditions
 - Example: A fixed mathematical calculation always produces the same answer
 - Random
 - Result is uncertain and may vary each time
 - Example: Throwing a die and observing the number
- Outcome
 - A single possible result from an experiment
 - Example: Getting a specific number such as 3 when rolling a die

Probability

Probability measures how likely an event is to happen. It assigns a value between zero and one.

- Interpretation

- Closer to zero means less likely
- Closer to one means more likely
- Notation
 - Probability of event A is written as $P(A)$
- Example
 - In a fair die, chance of getting any one number is equal since all outcomes are equally likely

Sample Space and Event

- Sample space
 - The complete set of all possible outcomes
 - Example: For a die $\rightarrow$ all numbers that can appear form the full set
- Event
 - A collection of outcomes we are interested in
 - Example: Getting an even number includes multiple outcomes from the sample space

Operations on Events

- Union of events
 - Represents occurrence of at least one event
 - Notation: $P(A \text{ or } B)$
 - Example
 - If one event is getting an even number and another is getting a number greater than four, the union includes outcomes satisfying either condition
- Intersection of events
 - Represents occurrence of both events together
 - Notation: $P(A \text{ and } B)$
 - Example
 - If one event is “even number” and another is “greater than two,” the intersection includes outcomes satisfying both

Special Types of Events

- Mutually exclusive events
 - Cannot occur at the same time
 - Their overlap is empty
 - Key idea
 - Probability of both happening together is zero
 - Example
 - Getting 2 and 5 in a single die roll
- Independent events
 - One event does not influence the other
 - Key idea
 - Combined probability equals product of individual probabilities
 - Example
 - Tossing a coin and rolling a die
- Dependent events
 - One event affects the outcome of another
 - Example
 - Drawing two balls from a bag without putting the first back

Certain and Impossible Events

- Certain event
 - Always occurs
 - Probability equals one
 - Example
 - Getting a number between 1 and 6 when rolling a standard die
- Impossible event
 - Cannot occur
 - Probability equals zero

- Example
 - Getting a number greater than 6 on a standard die

Exhaustive Events

- A group of events that together include every possible outcome
- At least one of them must happen
- Example
 - Getting an even number or an odd number on a die

Complement of an Event

The complement represents all outcomes where the event does not occur.

- Notation
 - Complement of A is written as A^c
- Rule
 - Probability of complement = one minus probability of the event
- Example
 - If the chance of rain is 0.3, then the chance of no rain is obtained by subtracting from one

Addition Rule

Used when finding the probability that at least one of multiple events occurs.

- General idea
 - Add individual probabilities but subtract overlap to avoid double counting
- Expression
 - $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
- Special case
 - If events cannot occur together, simply add their probabilities
- Example

- If two events have probabilities 0.4 and 0.5, and they overlap by 0.2, the final value is adjusted by subtracting the overlap

Multiplication Rule

Used when finding the probability of events occurring together.

- For independent events
 - Multiply their probabilities
 - Example
 - First event has probability 1 out of 6
 - Second event has probability 3 out of 6
 - Combined probability is product of the two
- For dependent events
 - Multiply probability of first event with conditional probability of second
 - Example
 - Drawing items without replacement reduces total possibilities for the next step
 - The second probability changes based on the first outcome

Conditional Probability

- Represents probability of one event given that another has already occurred
- Key idea
 - The total possible outcomes shrink based on the given condition
- Expression
 - Probability of A given B = probability of both A and B divided by probability of B
- Example
 - Finding probability of selecting a specific type of card when it is already known that the card belongs to a certain category